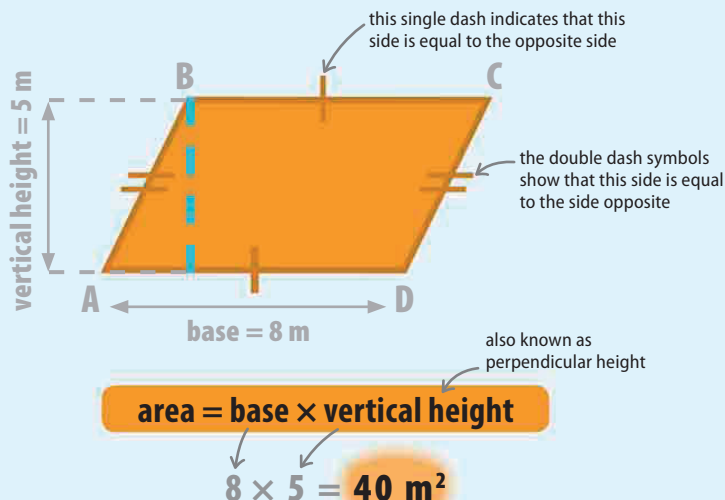


Finding the area of a parallelogram

Like the area of a rhombus, the area of a parallelogram is found by multiplying the length of its base by its vertical height.

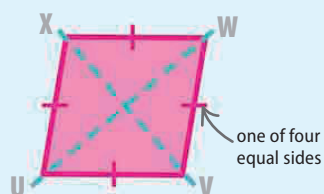
▷ Multiply base by vertical height

It is important to remember that the slanted side, AB, is not the vertical height. This formula only works if the vertical height is used.

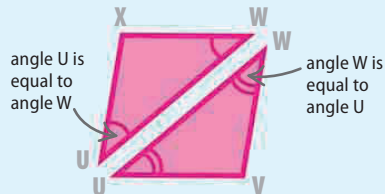


Proving the opposite angles of a rhombus are equal

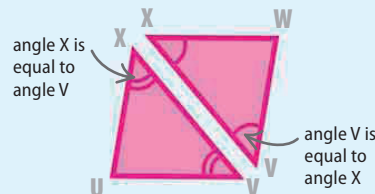
Creating two pairs of isosceles triangles by dividing a rhombus along two diagonals helps prove that the opposite angles of a rhombus are equal. An isosceles triangle has two equal sides and two equal angles.



All the sides of a rhombus are equal in length. To show this a dash is used on each side.



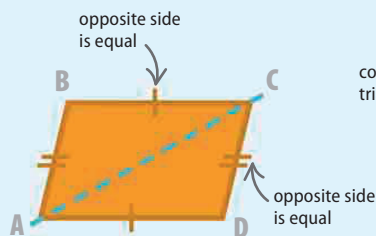
Divide the rhombus along a diagonal to create two isosceles triangles. Each triangle has a pair of equal angles.



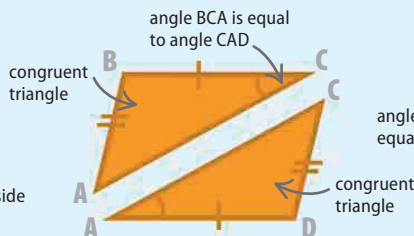
Dividing along the other diagonal creates another pair of isosceles triangles.

Proving the opposite sides of a parallelogram are parallel

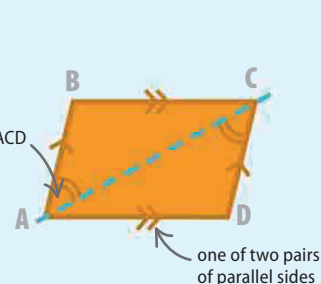
Creating a pair of congruent triangles by dividing a parallelogram along two diagonals helps prove that the opposite sides of a parallelogram are parallel. Congruent triangles are the same size and shape.



Opposites sides of a parallelogram are equal in length. To show this a dash and a double dash are used.



The triangles ABC and ADC are congruent. Angle BCA = CAD, and because these are alternate angles, BC is parallel to AD.



The triangles are congruent, so angle BAC = ACD; because these are alternate angles, DC is parallel to AB.